

Reg No.: \_\_\_\_\_

Name: \_\_\_\_\_

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**  
B.Tech Degree S7 (R, S) Examination November 2024 (2019 Scheme)

**Course Code: CST415**

**Course Name: INTRODUCTION TO MOBILE COMPUTING**

**Max. Marks: 100****Duration: 3 Hours**

**PART A**

*Answer all questions, each carries 3 marks.*

Marks

- |    |                                                                                                 |     |
|----|-------------------------------------------------------------------------------------------------|-----|
| 1  | Which are the different types of middleware components and gateways needed in mobile computing? | (3) |
| 2  | What are the main segments of a network?                                                        | (3) |
| 3  | Compare the Satellite systems-GEO, LEO and MEO                                                  | (3) |
| 4  | Explain in detail about IMEI, IMSI and MSISDN                                                   | (3) |
| 5  | How is billing and charging done in GPRS?                                                       | (3) |
| 6  | Describe the quality of service parameters for packet data networks.                            | (3) |
| 7  | What are advantages and disadvantages of wireless LAN?                                          | (3) |
| 8  | Define exposed terminal problem and hidden terminal problem.                                    | (3) |
| 9  | What are the different types of static and dynamic attacks on assets?                           | (3) |
| 10 | How do you ensure security in mobile computing environment through mobile VPN?                  | (3) |

**PART B**

*Answer any one full question from each module, each carries 14 marks.*

**Module I**

- |    |                                                                                                                                   |     |
|----|-----------------------------------------------------------------------------------------------------------------------------------|-----|
| 11 | a) What are the attributes needed to be considered for content adaptation and which are the major classifications of transcoding? | (9) |
|    | b) Describe the mobile computing functions.                                                                                       | (5) |

**OR**

- |    |                                                                                                  |     |
|----|--------------------------------------------------------------------------------------------------|-----|
| 12 | a) Explain seamless communication. What are aspects needed to be maintained in seamless roaming? | (7) |
|    | b) Describe in detail about client context manager                                               | (7) |

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**Module II**

- 13 a) Differentiate between Telecommunication networks and Internet? (4)  
b) Explain the working of GSM with the help of a neat diagram. (10)

**OR**

- 14 a) Define multiple accesses and its importance. Describe FDMA, TDMA, CDMA and SDMA with example (9)  
b) Draw a call flow diagram for a theatre ticket booking system. (5)

**Module III**

- 15 a) Describe the two methods used for accessing SMS bearer. (8)  
b) Explain various applications of GPRS in detail. (6)

**OR**

- 16 a) Explain the architecture of SMS in detail with the help of a diagram. (9)  
b) Describe briefly the data services in GPRS. (5)

**Module IV**

- 17 a) Briefly define how security is achieved in Wireless LAN. (7)  
b) Describe virtual carrier sense mechanism with suitable diagram. (7)

**OR**

- 18 a) Explain the HIPERLAN communication models with suitable diagrams. (6)  
b) How fragmentation and reassembling done in wireless LAN? Also explain different types of inter frame spaces. (8)

**Module V**

- 19 a) Explain the significance of Orthogonal Frequency Division Multiplexing. (6)  
b) Describe public key cryptography with the help of a diagram. (8)

**OR**

- 20 a) Given  $p = 3$ ,  $q = 11$  and  $e = 7$ . Find the value of  $d$  and also encrypt the message  $m = 2$  using RSA. (8)  
b) Explain the security frameworks for mobile computing. (6)

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